# Week 2 Notes

STAT 251 SECTION 01

# Describing and Visualizing Distributions

As we learned in the previous lecture, descriptive statistics is the first step to statistical analysis and involves the summary and organization of data. One of the main ways we can summarize data is to describe the characteristics of the **distribution** of the variable(s) in the data. A distribution tells us something about what kinds of values a variable can have and how often they occur.

## Data used this week

Each dataset below is a plain `.csv`, ready to read straight into R or to upload to the tools and calculators (https://stat-tools.jmks-homelab.dev/).

- Teen cell-phone survey (./Data/teens_cell_phone_focus_simulated.csv) — 743 responses (Figure 1 below).
- Old Faithful eruptions (./Data/old_faithful.csv) — 272 eruptions, duration and waiting time.
- MPG and engine cylinders (./Data/carsdata.csv) — 32 car models, used for the dot plot examples in class.
- Consider the following example: Are teens distracted by their cell phones? A study conducted by the Pew Research center surveyed 743 U.S teenagers ages 13-17 to understand how cell phones in class impacted their ability to concentrate. Each student was asked to rate the impact of cell phone use on their concentration based on a Likert scale. The data and a plot of the distribution of survey responses are given below

Survey data layout: the first four teens and the last, of 743.

| Student | Age | Response |
|---|---|---|
| 1 | 13 | Never |
| 2 | 13 | Sometimes |
| 3 | 15 | Never |

| Student | Age | Response |
|---|---|---|
| 4 | 17 | Often |
| ⋮ | ⋮ | ⋮ |
| 743 | 16 | Rarely |

- What type of variable is *Age*?
- What type of variable is *Response*?

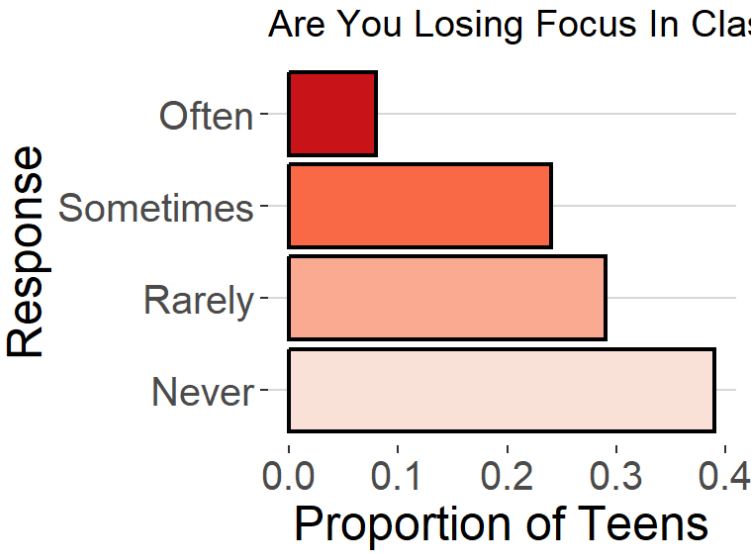



Figure 1. Distribution of student responses to the survey question "Are you losing focus in class by checking your cell phone?"

Download the data behind Figure 1 (./Data/teens_cell_phone_focus_simulated.csv) — one row per student. Pew published only the four percentages, so the individual rows are simulated; the response counts (289 Never, 216 Rarely, 178 Sometimes, 60 Often, n = 743) are the published ones and reproduce Figure 1 exactly.

- Looking at the distribution of survey responses above, would you say that most teens feel they are distracted by their cell phones?

## Distributions

Formally, a **distribution** is a function that gives (a) the possible values of a variable and (b) how often each value occurs. The *how often* is usually described using *frequency* or *relative frequency*.

- The **frequency** of a given value is simply a count of the number of times that the given values occurs
- The **relative frequency** of a given value is the proportion of times that the given value occurs
- We can also use the *cumulative relative frequency* to describe how often a variable occurs. The **cumulative relative frequency** is the proportion of values that are *less than or equal to* a given value in a distribution.
    - **Note**: As we said with the example about distracted teens, we can talk about the distribution of a categorical variable as well, however we *may not use* the cumulative relative frequency if there is no natural order to the values (i.e qualitative - nominal).

Consider the following hypothetical set of $10$ exam scores (out of 10 points) for students in a statistics course

$$\{5, 6, 6, 7, 7, 7, 8, 9, 9, 10\}$$

- the possible test scores are $5, 6, 7, 8, 9,$ and $10$. Lets try computing the frequency, relative frequency, and cumulative relative frequency for students who score a 6 on the exam.
    - The frequency for an exam score of $6$ is $freq(6) = 2$ because $2$ of the $10$ students had an exam score of $6$.
    - Similarly, the relative frequency for an exam score of $6$ is $2/10 = 0.2$.
    - The cumulative relative frequency for an exam score of $6$ will be the total number of values in the data that are less than or equal to the value $6$. This will be the frequency of $5$'s plus the frequency of $6$'s which will be $(1 + 2)/10 = 3/10$ or $0.3$. Alternatively we could simply add the relative frequency for an exam score of $5$ to the relative frequency for an exam score of $6$: $0.1 + 0.2 = 0.3$

Frequency table for the ten exam scores.

| Score | Frequency | Relative Frequency | Cumulative Relative Frequency |
|---|---|---|---|
| 5 | 1 | 0.1 | 0.1 |
| 6 | 2 | 0.2 | 0.3 |
| 7 | 3 | 0.3 | 0.6 |
| 8 | 1 | 0.1 | 0.7 |
| 9 | 2 | 0.2 | 0.9 |

| Score | Frequency | Relative Frequency | Cumulative Relative Frequency |
|---|---|---|---|
| 10 | 1 | 0.1 | 1.0 |

- The table above is called a **frequency table**. It is a tabular display of the distribution of a variable.

A frequency table is only one way of describing a distribution, and it is the plainest. The rest of these notes introduce the other ways: how to build a frequency table when the variable is continuous, what features of a distribution we look for, and which graph suits which kind of variable.

## Frequency tables for continuous variables

Every frequency table above listed the possible values of the variable in its first column and counted them in the rest. That works whenever the values can be listed.

We can create a frequency table for any type of variable. However, when dealing with quantitative continuous variables we cannot list all possible values because the number of possible values is infinite. Therefore, we usually have to make a decision about how we want to group values of the variable, a process called binning. **Binning** is a way to group numbers of more-or-less continuous values into a smaller number of "bins". These bins represent a collection of values that fall into a certain range and make listing the possible values a lot easier.

Consider the following trivial example using the quantitative continuous variable

$$X = \{1.23,\ 1.38,\ 1.76,\ 2.60,\ 2.65,\ 2.79,\ 2.89,\ 3.12,\ 3.15,\ 3.18,\ 3.65,\ 3.77,\ 3.87,\ 4.12,\ 4.41\}$$

Each observation of the variable $X$ is measured with a precision of two decimal places, and no two observations in the sample are identical. As a result if we simply listed the possible values we would have $15$ unique values each with a frequency of $1$ which would produce a frequency table that wasn't very informative. We can, instead, use binning to extract more meaningful information about the distribution of $X$. Now consider the frequency table using binning:

Frequency table for X using seven bins of width 0.5.

| Bin | Frequency | Relative Frequency | Cumulative Relative Frequency |
|---|---|---|---|
| $1 < X \leq 1.5$ | 2 | 0.13 | 0.13 |
| $1.5 < X \leq 2$ | 1 | 0.07 | 0.20 |
| $2 < X \leq 2.5$ | 0 | 0.00 | 0.20 |

| Bin | Frequency | Relative Frequency | Cumulative Relative Frequency |
|---|---|---|---|
| $2.5 < X \leq 3$ | 4 | 0.27 | 0.47 |
| $3 < X \leq 3.5$ | 3 | 0.20 | 0.67 |
| $3.5 < X \leq 4$ | 3 | 0.20 | 0.87 |
| $4 < X \leq 4.5$ | 2 | 0.13 | 1.00 |

The following is another example using $n = 272$ observations collected on the waiting time between eruptions and the duration of the eruption of the Old Faithful geyser in Yellowstone National Park, Wyoming, USA.

Download the Old Faithful data (./Data/old_faithful.csv) (all 272 eruptions).

The first eight Old Faithful eruptions, of 272.

| Duration (min) | Waiting Time (min) |
|---|---|
| 3.6 | 79 |
| 1.8 | 54 |
| 3.333 | 74 |
| 2.283 | 62 |
| 4.533 | 85 |
| 2.883 | 55 |
| 4.7 | 88 |
| 3.6 | 85 |
| ⋮ | ⋮ |

Consider a frequency table to describe the distribution of the variable *waiting time (min)*:

Frequency table for Old Faithful waiting time using twelve bins of width 5 minutes.

| Waiting Time | Frequency | Relative Frequency | Cumulative Relative Frequency |
|---|---|---|---|
| $40 < X \leq 45$ | 4 | 0.015 | 0.015 |

| Waiting Time | Frequency | Relative Frequency | Cumulative Relative Frequency |
|---|---|---|---|
| $45 < X \leq 50$ | 22 | 0.081 | 0.096 |
| $50 < X \leq 55$ | 33 | 0.121 | 0.217 |
| $55 < X \leq 60$ | 24 | 0.088 | 0.305 |
| $60 < X \leq 65$ | 14 | 0.051 | 0.357 |
| $65 < X \leq 70$ | 10 | 0.037 | 0.393 |
| $70 < X \leq 75$ | 27 | 0.099 | 0.493 |
| $75 < X \leq 80$ | 54 | 0.199 | 0.691 |
| $80 < X \leq 85$ | 55 | 0.202 | 0.893 |
| $85 < X \leq 90$ | 23 | 0.085 | 0.978 |
| $90 < X \leq 95$ | 5 | 0.018 | 0.996 |
| $95 < X \leq 100$ | 1 | 0.004 | 1.000 |

- How long does a visitor typically have to wait before seeing old faithful erupt?
- Is there more than one "typical" value in this distribution?

**IMPORTANT**: one thing to consider is that grouping continuous data into discrete intervals leads to a loss of precision. Sometimes this can result in the loss of valuable information present in the original data. In addition, how we choose to bin the data (i.e how wide we make the intervals) may lead to different, sometimes erroneous, conclusions about the distribution and patterns within the data.

Consider the frequency table for the *waiting time (min)* with only two bins

The same waiting times collapsed into only two bins of width 30 minutes.

| Waiting Time | Frequency | Relative Frequency | Cumulative Relative Frequency |
|---|---|---|---|
| $40 < X \leq 70$ | 107 | 0.393 | 0.393 |
| $70 < X \leq 100$ | 165 | 0.607 | 1.000 |

- How does using only two bins/intervals alter your conclusion regarding the typical time a visitor must wait before witnessing Old Faithful's eruption?

## Features of a distribution

Descriptive statistics or graphical summaries can tell us a lot about the variables in our data. Graphs help us visualize trends in the data, identify problematic observations, and can help inform the types of analyses we should use. We may be interested in certain features related to the distribution(s) of the variable(s) in our data, such as the *shape, center, and variability*.

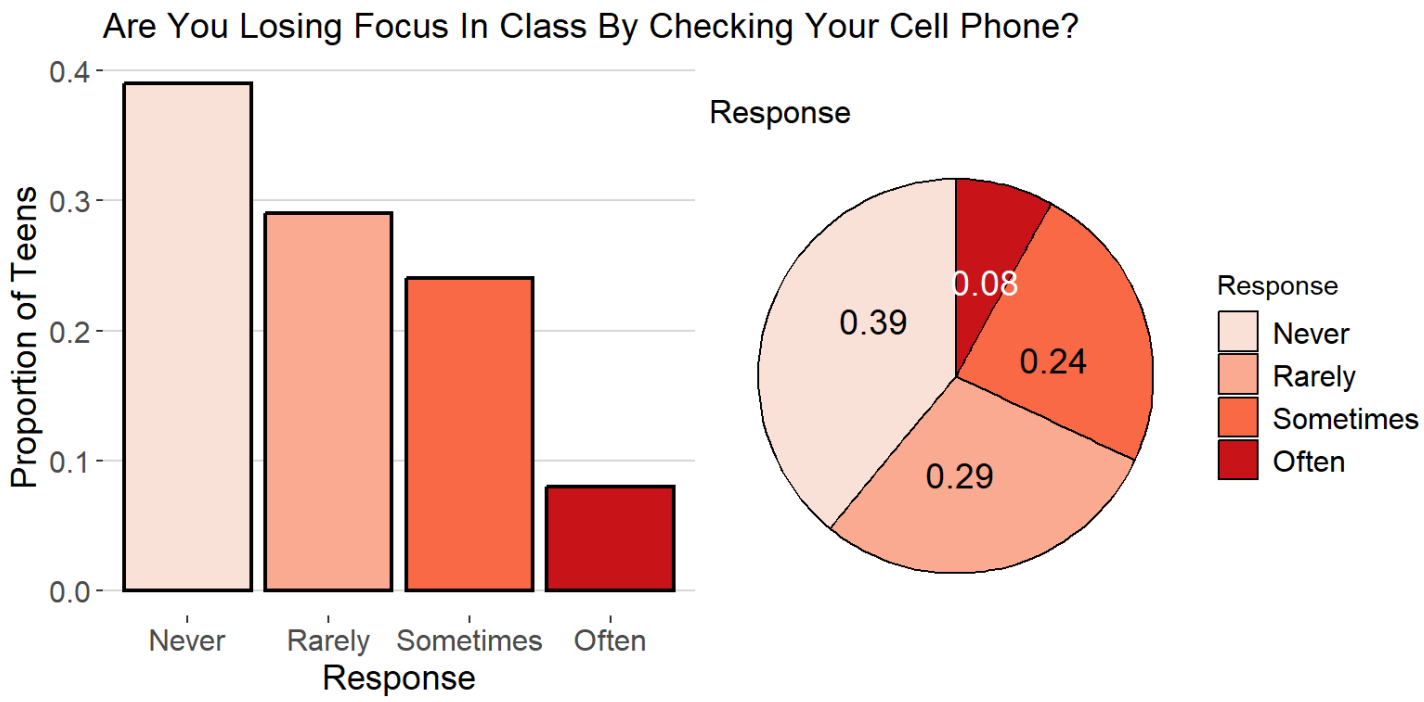
- **shape** refers to how observations cluster across the range of a variable. Certain aspects to note would be where observations are densely packed (values with high frequencies) and where observations are sparser (values with low frequencies). This can give us a feel for where a typical value would appear and what values may be considered "rare" or "extreme".
- **Center** refers to the "middle point" on the distribution and also supplies information about where a typical value falls.
- **Variability** refers to how tightly observations cluster around the center of a distribution. When values are tightly packed they are said to have *low variability* vs when they are spread out they are said to have *high variability*
- **modal category** is a feature of qualitative data only and refers to the most frequent category of a qualitative variable.

## Graphs for qualitative data

There are several graphs that can be used to display qualitative data. The most common are bar graphs and pie charts.

- A **Bar graph** is simple and widely used plot for displaying the distribution of categorical variables. Each category is represented by a bar, and the height of the bar corresponds to the frequency or proportion of observations in that category. When the bars are sorted in decreasing frequency it is called a **Pareto chart**.
- A **Pie chart** is useful for illustrating the proportional contribution of each category to the whole. Beyond this, it is not a particularly useful plot when describing the features of a distribution.
- Ex. Recall the Pew Research survey above, asking teens to rate how distracted they are by their cell phones. The following are a bar graph and pie chart of student responses

Download the response data (./Data/teens_cell_phone_focus_simulated.csv) (simulated student-level rows reproducing the published Pew counts).




- what is the modal category of the variable *Response*?

## Graphs for quantitative data

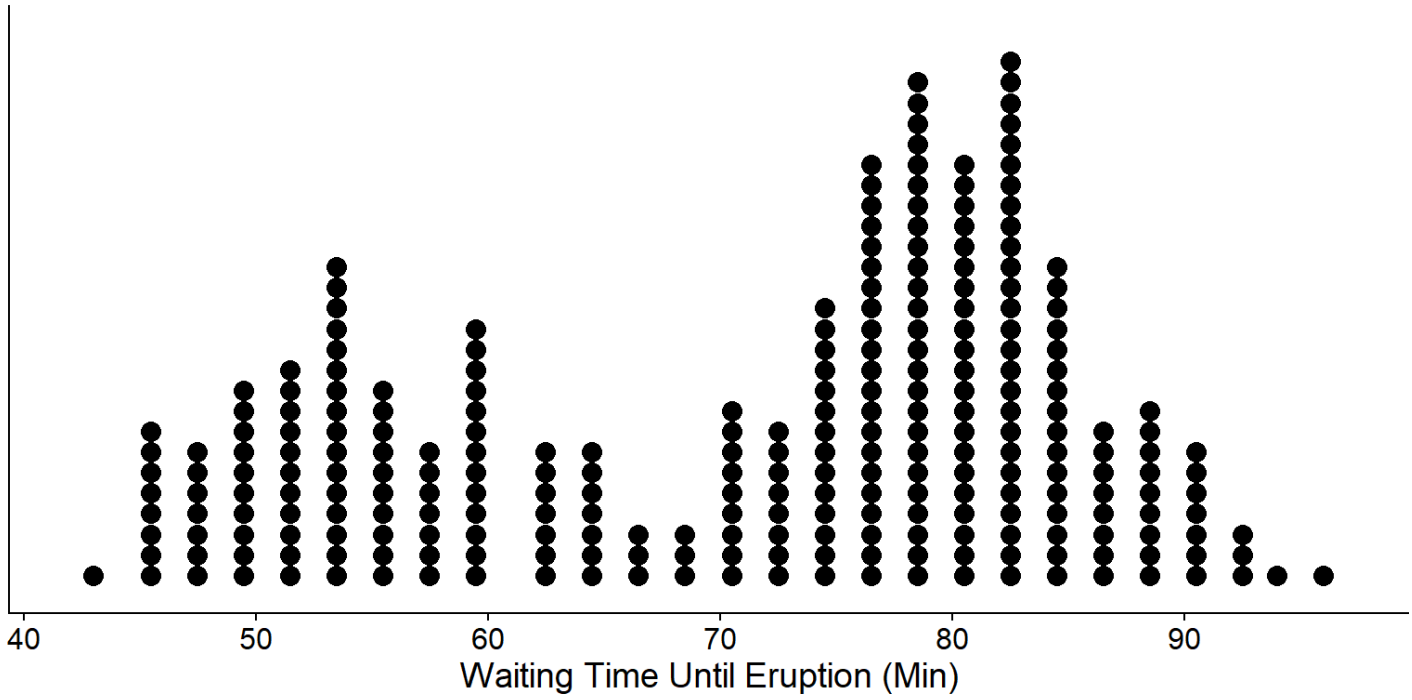
There are also many plots that can be used with quantitative data. Some of the most commonly used graphs are listed below.

In class we build these plots from the MPG and engine-cylinder data for 32 car models (./Data/carsdata.csv).

- A **dot plot** shows a dot for each observation placed just above the value on the number line for that observation. To construct a dot plot:
    - 1. Draw a horizontal line, label it with the name of variable, and mark regular values of the variable on the line.
    - 2. For each observation, place a dot above its value on the number line.
- A **stem and leaf plot** is similar to a dot plot in that it displays individual observations of a variable. Each observation in the plot consists of a stem and a leaf. The *stem* is all of the digits except for the last digit. The *leaf* is the last digit. For example, take the observation $12.3$ the stem for this observation is $12$ and the leaf is $3$. To construct a stem and leaf plot:
    - 1. Draw a table with two columns and label the left column “Stem” and the right column “leaf”.

- 2. Next, sort the observations from smallest to largest and place the stems in the left column of your table in increasing order.
- 3. in the right column, fill in the leaves of each stem in increasing order.

Example: A dot plot constructed for the $n = 272$ observations collected on the waiting time between eruptions and the duration of the eruption of the Old Faithful geyser in Yellowstone National Park, Wyoming, USA.




- Think about the shape of this distribution and how it relates to the time that a typical visitor can expect to wait before the geyser erupts. Is the result consistent with the frequency table we constructed earlier?

Example: A stem and leaf plot constructed for the $n = 272$ observations collected on the waiting time between eruptions and the duration of the eruption of the Old Faithful geyser in Yellowstone National Park, Wyoming, USA.

Read the plot below one row at a time. The number to the left of the bar is the stem, here the tens digit of the waiting time; each digit to the right of the bar is one leaf, the units digit of a single eruption. A row reading `5 | 0011` would stand for four eruptions, at 50, 50, 51 and 51 minutes.

Note that R has *split* each stem across two rows, so every tens digit appears twice: the first row collects the leaves 0 through 4, the second the leaves 5 through 9. The two rows beginning `5` therefore cover 50–54 minutes and 55–59 minutes respectively.

The longer a row is, the more eruptions fell in that five-minute span. That is what makes the shape audible as well as visible: the rows grow long in the low fifties, shrink to their shortest in the middle sixties, then grow longest of all in the high seventies and low eighties — the same two clusters, and

the same gap between them, that the dot plot showed.

```
##
##   The decimal point is 1 digit(s) to the right of the |
##
##   4 | 3
##   4 | 55566666777788899999
##   5 | 000001111112222233333334444444444
##   5 | 555555666677788889999999
##   6 | 00000022223334444
##   6 | 555667899
##   7 | 000011111233333334444444
##   7 | 55555555666666666777777777777888888888888889999999999
##   8 | 000000001111111111111222222222222333333333333334444444444
##   8 | 55555566666677888888999
##   9 | 00000012334
##   9 | 6
```

Both dot plots and stem and leaf plots are excellent choices for discrete variables with a small to moderate number of observations. However, when a variable consists of many observations, has low variability, or is continuous, it can be difficult to construct these two plots.

- When dealing with continuous data, it is often necessary to round observations to the nearest tenth before constructing either plot.

The following are alternatives that work for variables of any number of observations.

## More graphs for quantitative data

- A **Histogram** is a specialized type of bar graph. It uses bars to portray the frequencies or relative frequencies of the possible outcomes for a quantitative variable of either type. The histogram uses binning to condense the view of a quantitative variable into a smaller set of intervals which makes it particularly useful when the number of observations is large. The width of the bins is arbitrary but all bins should be the same length and cover all possible values of the data. The steps for constructing a histogram are:
    - 1. Divide the data into non-overlapping intervals (bins) of equal length.
    - 2. Compute the frequency of each interval (this is the same as constructing the frequency table for the chosen bins)

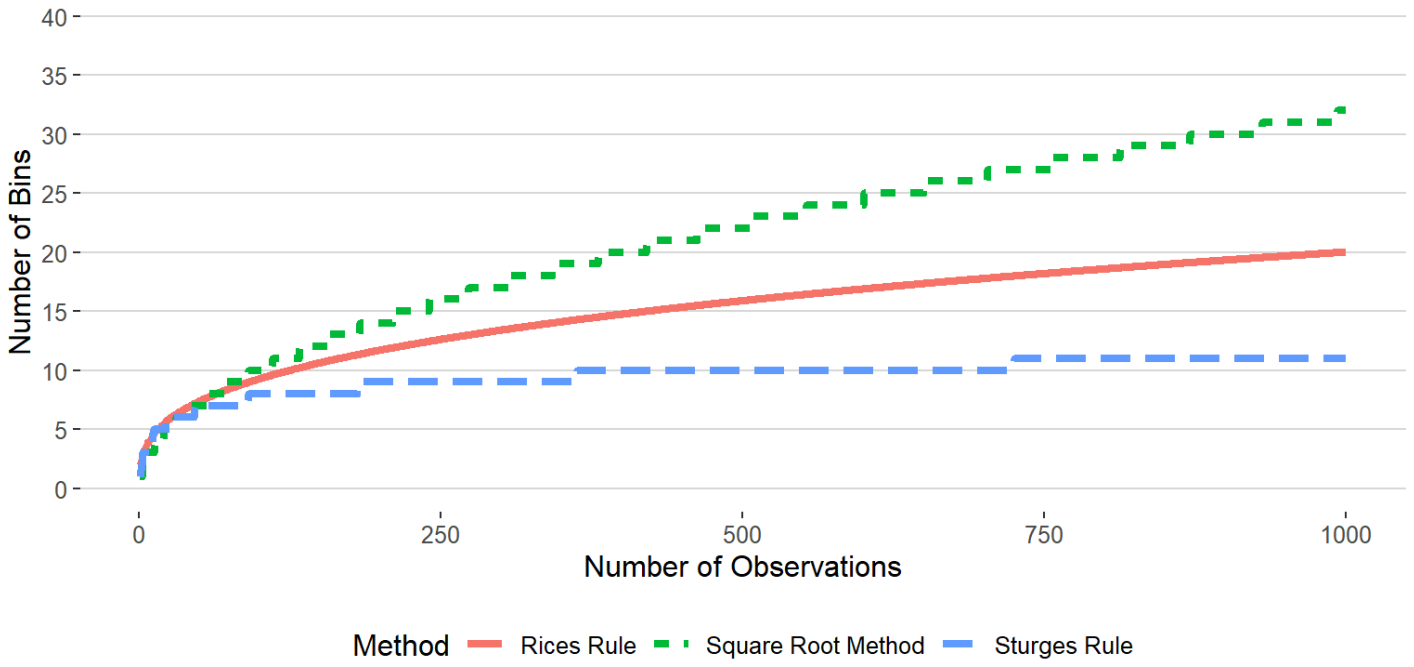
    - 3. Label the *x*-axis with the midpoints or endpoints of each interval.
    - 4. Draw a bar over each value or interval with height equal to its frequency or relative frequency
- In general, you can choose the number of bins $k$ you wish to use and set the bin width $w$ accordingly or alternatively set the bin width you want and use the corresponding number of bins.
    - formula for choosing the number of bins based on bin width $w$

$$k = \frac{\max\ x - \min x}{w}$$

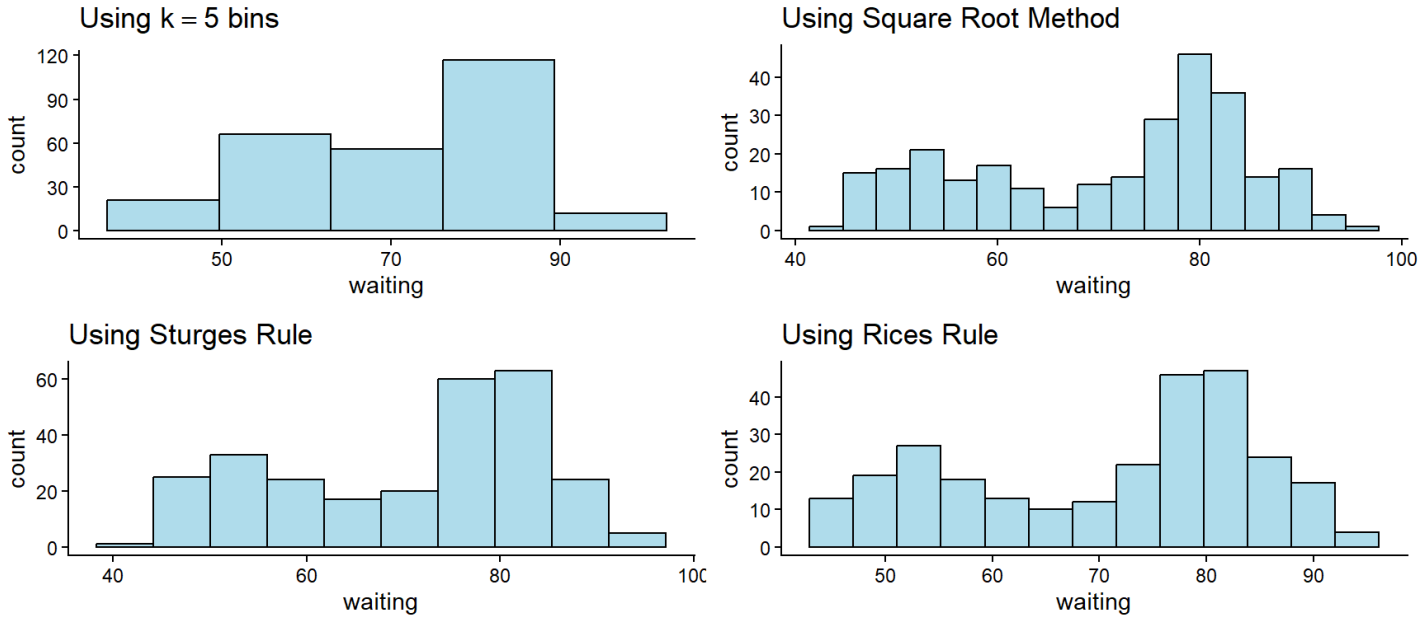
    - formula for choosing the width based on the number of bins $k$

$$w = \frac{\max x - \min x}{k}$$

  where $\min$ and $\max$ denote the smallest and largest values of the variable $x$, respectively.
- How many bins should I use? This is actually a question that has received a lot of attention in the scientific literature. Typically, you want to be sure that you don't use "too few" bins as this can cause misleading conclusions about the features of the distribution. There are several rules that are based on the coverage and number of observations of the variable:
    - **square root method** is defined as $k = \text{round}(\sqrt{n})$. This is a fairly easy and safe rule for choosing the number of bins to use. However, for large sample sizes it can lead to a very large number of bins.
    - **Sturges Rule** is defined as $k = \text{round}(\log_2 n) + 1$. This method is not great for a small number of observations (i.e $n < 30$)
    - **Rices Rule** is defined as $k = 2\sqrt[3]{n}$. This rule is a nice intermediate between Sturges Rule and the square root method.




Consider the following example of histograms constructed for the $n = 272$ observations collected on the waiting time between eruptions and the duration of the eruption of the Old Faithful geyser in Yellowstone National Park, Wyoming, USA.




**Try it out**: Construct a histogram for the following 10 observations of the variable $X$

$$X = \{-1.49, -0.65, -0.6, -0.54, -0.45, 0.01, 0.17, 0.27, 0.51, 1.34\}$$

using $k = 4$ bins/intervals

- In this example, we are given the desired number of bins, so we can use our equation to find the width of the bins $w$.

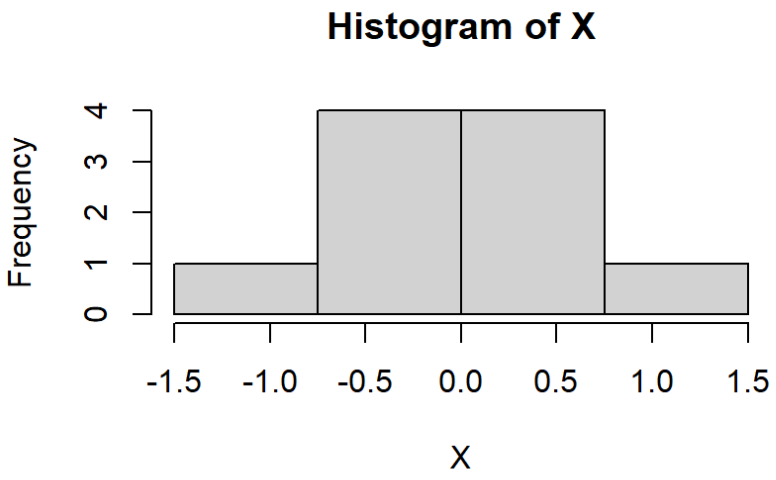
$$w = \frac{1.34 - (-1.49)}{4} = \frac{2.83}{4} = 0.7075 \approx \frac{3}{4} = 0.75$$

- now that we know about how wide each interval should be we can easily comupute the intervals. First, let $a$ be the minimum value of $X$ and $b$ be the maximum value such that $\min(X) = a, \max(X) = b$. The first interval is defined as $a < X \leq a + w$ which gives $-1.5 < X \leq -0.75$. The second interval is given by $-0.75 < X \leq -0.75 + w$ or $-0.75 < X \leq 0$. Continuing this pattern gives the remaining bins yields the following frequency table:

Frequency table for X using four bins of width 0.75.

| X | Frequency | Relative Frequency | Cumulative Relative Frequency |
|---|---|---|---|
| $-1.5 < X \leq -0.75$ | 1 | 0.1 | 0.1 |
| $-0.75 < X \leq 0$ | 4 | 0.4 | 0.5 |
| $0 < X \leq 0.75$ | 4 | 0.4 | 0.9 |
| $0.75 < X \leq 1.5$ | 1 | 0.1 | 1.0 |

- From the frequency table above, we can now easily construct the histogram of $X$ by simply drawing the appropriate bars over each interval/bin:




If you'd like to read more about scaling interval widths you can check out these references:

Sturges, Herbert A."The choice of a class interval." Journal of the american statistical association 21.153 (1926): 65-66. (https://scholar.archive.org/work/l44cm5gq65f4rojkkaey6cl5bq/access/wayback/http://www.esalq.usp.br/departamentos/lce/arquivos/aulas/2013/LCE0216/Sturges1926.pdf)

Lane, David. Online statistics education: A multimedia course of study. Association for the Advancement of Computing in Education (AACE), 2003. – Chapter 2 "Graphing Distributions (https://open.umn.edu/opentextbooks/formats/651)

## Describing the shape of a distribution

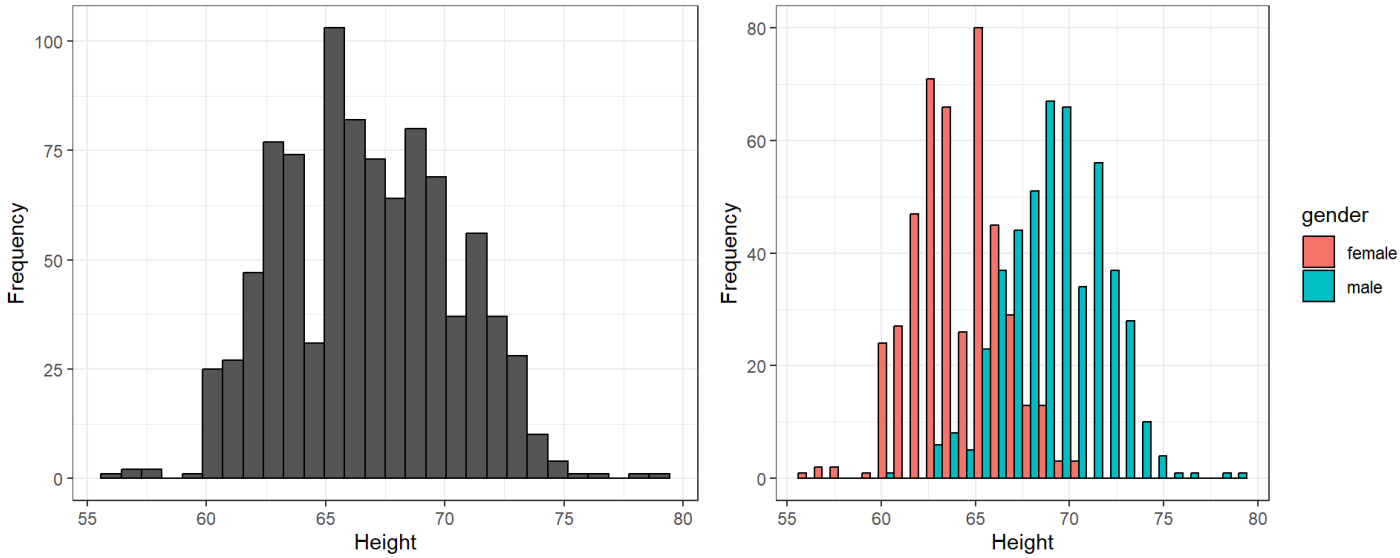
The purpose of a graph is to help us understand the data. Once the distribution of variable is plotted, you must ask yourself "what do I see?". This is when we look for the features of a distribution such as shape, center and spread. This helps us to identify interesting patterns or potential **outliers** - individual data points that lie outside the normal pattern of the distribution.

- One of the first things we can look for are peaks of the distribution. A peak indicates values that are more likely to occur called the **mode**. A distribution can have no peaks, one peak, or multiple peaks. A distribution with one peak is said to be **unimodal**, a distribution with two peaks is **bimodal** and a distribution with multiple peaks is **multimodal**.
    - bimodal or multimodal distributions typically indicate underlying population structure in the data. Take for example the following height data from Sir Francis Galton (1822–1911) who was an English statistician. Galton was interested in the relationship between the height of parents and their offspring. He accumulated the following observations of the height of 934 children from 205 families in 1886.

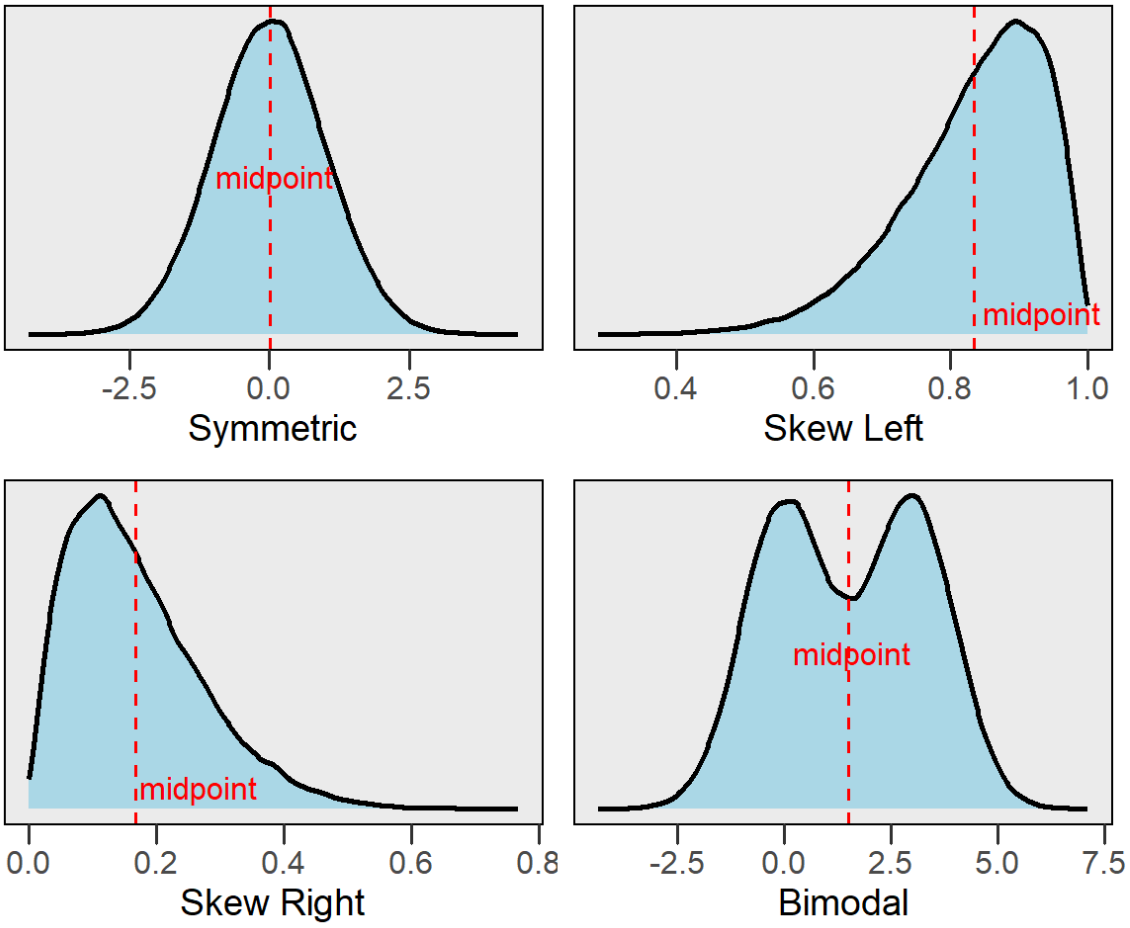
| family | father | mother | midparentHeight | children | childNum | gender | childHeight |
|---|---|---|---|---|---|---|---|
| 001 | 78.5 | 67 | 75.43 | 4 | 1 | male | 73.2 |
| 001 | 78.5 | 67 | 75.43 | 4 | 2 | female | 69.2 |
| 001 | 78.5 | 67 | 75.43 | 4 | 3 | female | 69 |
| 001 | 78.5 | 67 | 75.43 | 4 | 4 | female | 69 |
| 002 | 75.5 | 66.5 | 73.66 | 4 | 1 | male | 73.5 |
| 002 | 75.5 | 66.5 | 73.66 | 4 | 2 | male | 72.5 |
| 002 | 75.5 | 66.5 | 73.66 | 4 | 3 | female | 65.5 |
| 002 | 75.5 | 66.5 | 73.66 | 4 | 4 | female | 65.5 |
| 003 | 75 | 64 | 72.06 | 2 | 1 | male | 71 |
| 003 | 75 | 64 | 72.06 | 2 | 2 | female | 68 |

| family | father | mother | midparentHeight | children | childNum | gender | childHeight |
|---|---|---|---|---|---|---|---|
| NA | ⋮ | ⋮ | ⋮ | ⋮ | ⋮ | NA | ⋮ |
| 204 | 62.5 | 63 | 65.27 | 2 | 2 | female | 57 |

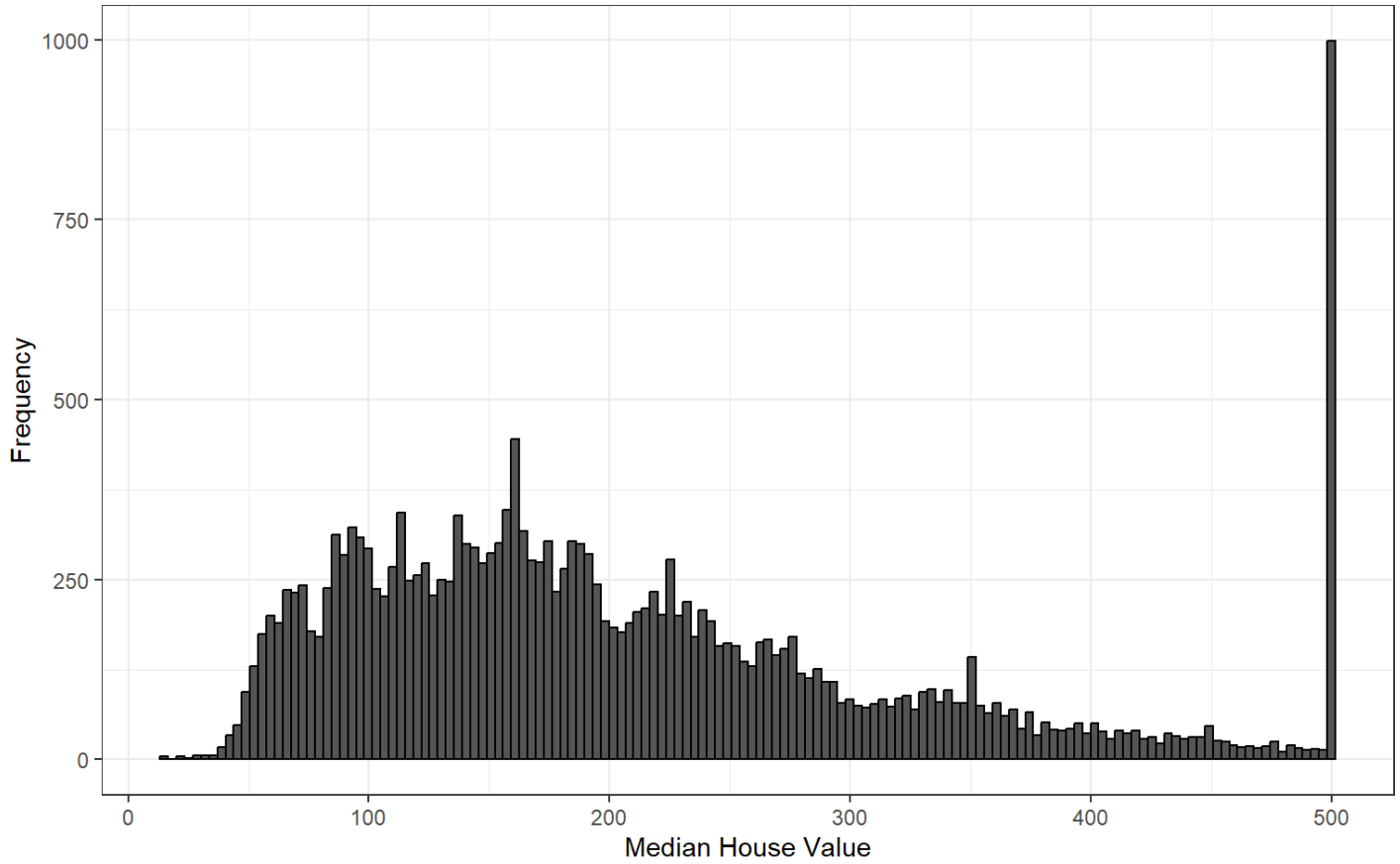
Now consider the histogram for child height:



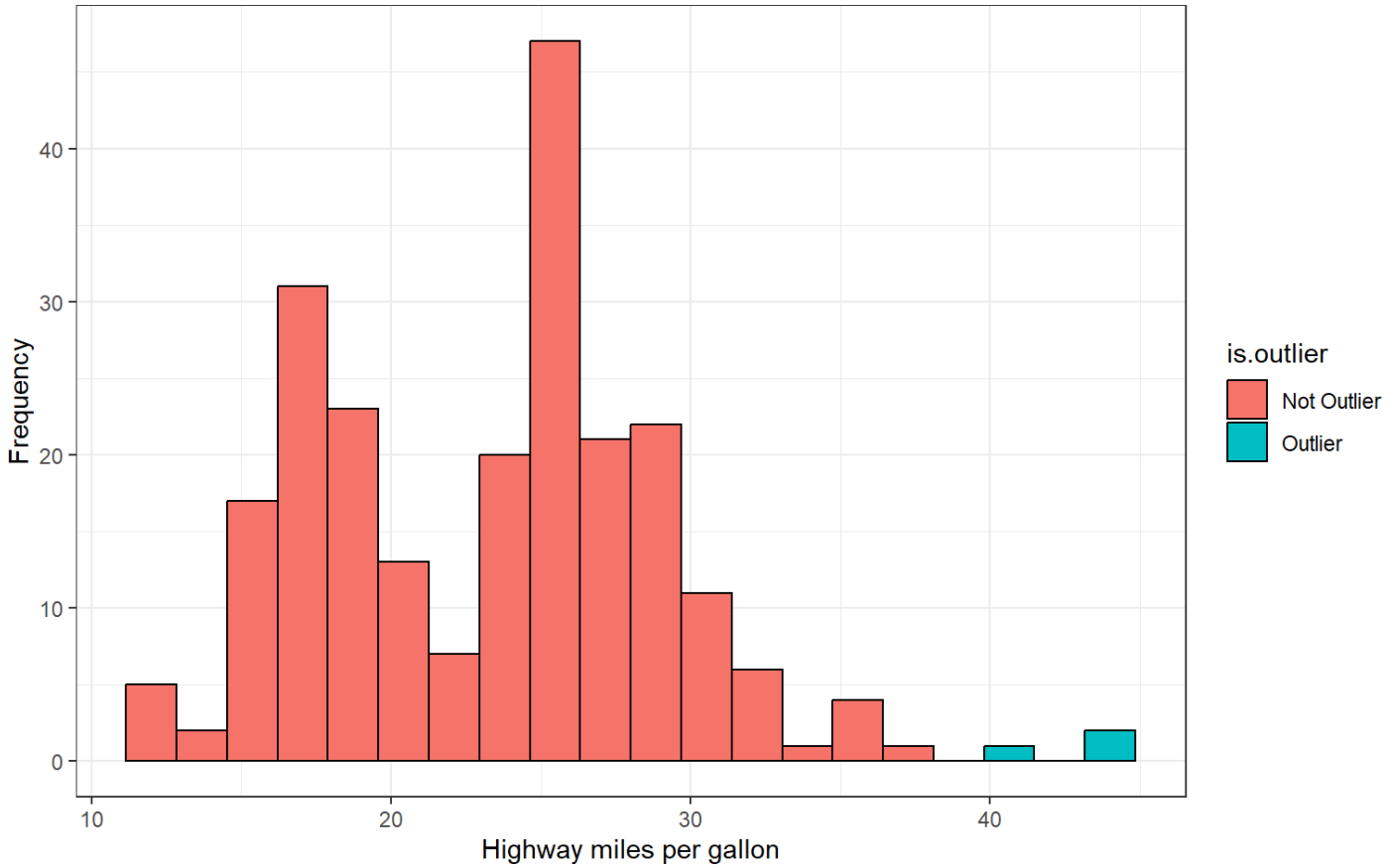

- We can see from the histogram on the left that there are two clearly visible peaks. By coloring and segregating the bars according to gender we produce the histogram on the right which shows that the two peaks correspond to the gender of the children with males being about 5 inches taller on average.
- Another question we may ask about the shape of a distribution is whether the distribution is symmetric or if it is skewed. A distribution is **symmetric** if the values that are smaller and larger than its midpoint are mirror images. A distribution is **skewed right** if the values larger than the midpoint form a long tail. Conversely, a distribution is **skewed left** if the values less than the midpoint form a long tail to the left




Many types of variables have distributions with predictable shapes. For example, variables that represent measurements of size from the same species and sex such as height, length, and age will have symmetric distributions. Variables related to money or finance on the other hand will often have skewed distributions. Skewness arises when there is a strict boundary on the values of a variable. For example, housing prices are always positive numbers and many houses will be a similar price while some houses may be really expensive. For example, consider the distribution of housing prices from the 1990 california housing census data downloaded from Kaggle (https://www.kaggle.com/datasets/camnugent/california-housing-prices?resource=download)




Lastly, one of the most important things a graphical summary of a variable can show is **outliers**. As previously noted, these are extreme values that fall far away from the normal pattern of the data. An outlier may be a result of measurement error or it can represent a real datapoint. Either way, outliers can strongly influence any analyses we conduct later on during statistical inference and are crucial to be aware of. Consider the following histogram of fuel efficiency (measured in miles per gallon) of several different car brands between 1999 and 2008 obtained from https://fueleconomy.gov/ (https://fueleconomy.gov/)




- How would you describe the shape of this distribution?
- From the histogram we can see there are two bars that fall far to the right of the rest of the distribution. The observations comprising these bars constitute potential outliers. In fact, these three observations belong to the 1999 volkswagen beetle and jetta which was extremely fuel efficient for its time

## Describing distributions with numbers

Any description of the distribution of a variable is not complete without some numbers describing its *center* and *spread*. Previously, we learned how to describe shape by visually inspecting a graph of the data such as a histogram, stem plot, or dot plot. Now we will learn some statistics we can calculate that will tell us about spread and center. It's important to note that we can calculate these descriptive statistics for any quantitative variables but NOT for qualitative variables. We will start by talking about statistics for quantifying the center of a distribution. Two measures of center are the *mean* and the *median*

## Measures of center

The arithmetic **mean** (denoted $\bar{x}$) is the average value of a set of observations and tells us the center of mass or balancing point of a distribution

$$\bar{x} = \sum_{i=1}^{n} \frac{x_i}{n} = \frac{x_1 + x_2 + \cdots + x_n}{n}$$

- the symbol $\Sigma$ means to "add up a set of values". Therefore, the mean is the sum of all observations in the sample (or population) and divide the sum by the number of observations
- the mean is usually not equal to any of the values observed in the sample
- The mean is highly influenced by outliers. Outliers pull the value of the mean in their direction.

The other measure of center is the **median** or middle value. To calculate the median, we must first arrange the data in order from least to greatest. If the number of observations $n$ is odd then the median is exactly the middle value. If the number of observations $n$ is even, then the median is the "average" of the two values in the middle.

- The median is resistant to outliers because it does not depend on their value like the mean does.

Example: consider the following 8 observations of a quantiative variable $X$:

$$X = \{1, 3, 5, 5, 6, 7, 7, 8\}$$

Compute the mean and median of $X$

$$\bar{x} = \frac{1 + 3 + 5 + 5 + 6 + 7 + 7 + 8}{8} = 5.25$$

$$\text{median} = \frac{5 + 6}{2} = 5.5$$

Now consider a the mean and median when we include the outlier observation $X = 32$ and note it's effect each statistic:

$$X = \{1, 3, 5, 5, 6, 7, 7, 8, 32\}$$

The mean becomes

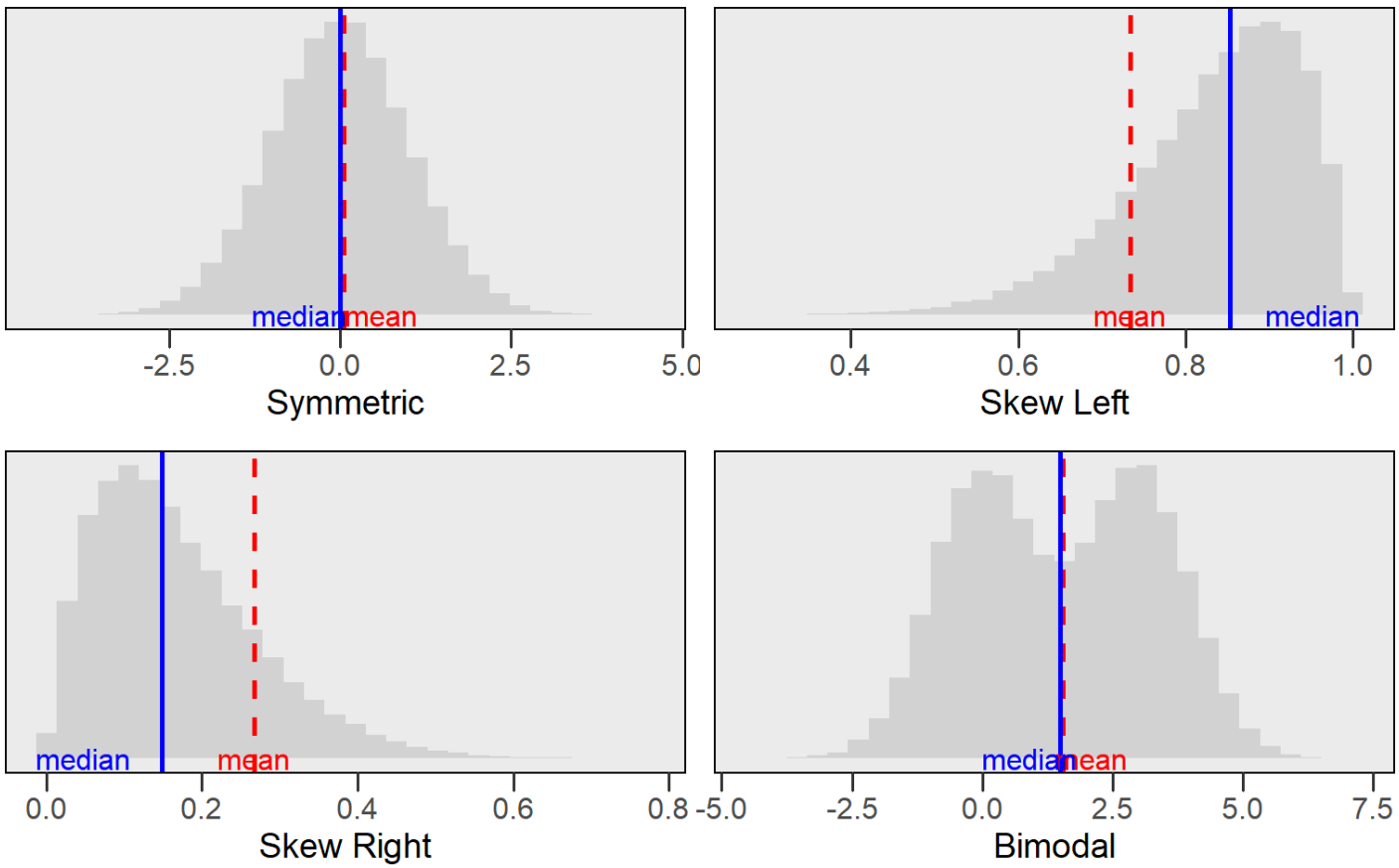
$$\bar{x} = \frac{1 + 3 + 5 + 5 + 6 + 7 + 7 + 8 + 32}{9} = 8.22$$

The median becomes

$$\text{median} = 6$$

Notice that in the first set of calculations the mean was less than the median. Now, because of the presence of the outlier, the mean has surpassed the value of the median because the “weight” that the outlier carries in terms of its value.

- This example illustrates how the mean and median behave depending the shape of a distribution. Distributions with skew will pull the value of the mean in the direction of the tail while the median will typically lie between the mode and the mean of the distribution




## An alternative formula for the mean

We can also compute the mean using the frequency or relative frequency using the following formulas

Using the frequency

$$\bar{x} = \sum_{x} \frac{x \times \text{freq}(x)}{n}$$

Using the relative frequency

$$\bar{x} = \sum_{x} x \times rf(x)$$

- where $\sum_x$ means to sum over all values of x and $\boldsymbol{freq(x)}$ and $\boldsymbol{rf(x)}$ denote the frequency and relative frequency of $\boldsymbol{x}$ respectively.

Example: Sir Ronald Fisher's famous iris data set gives the measurements in centimeters of the variables sepal length and width and petal length and width, respectively, for 50 flowers from each of 3 species of iris. Consider the following frequency table of petal lengths rounded to the nearest whole centimeter

| Petal Length (cm) | Frequency | Relative Frequency | Cumulative Relative Frequency |
|---|---|---|---|
| 1 | 24 | 0.16 | 0.16 |
| 2 | 26 | 0.17 | 0.33 |
| 5 | 3 | 0.02 | 0.35 |
| 4 | 34 | 0.23 | 0.58 |
| 3 | 35 | 0.23 | 0.81 |
| 6 | 24 | 0.16 | 0.97 |
| 7 | 4 | 0.03 | 1.00 |

We can compute the mean petal length using either of our formulas from above:

- Using frequency

$$\bar{x} = \frac{1(24) + 2(26) + 3(3) + 4(34) + 5(35) + 6(24) + 7(4)}{150} \approx 3.79$$

- Using the relative frequency

$$\bar{x} = 1(0.16) + 2(0.17) + 3(0.02) + 4(0.23) + 5(0.23) + 6(0.16) + 7(0.03) \approx 3.79$$

## Measures of location: the mode

We have several numerical quantities that we can compute from a distribution that tell us where certain values fall in the distribution. We call these measures of location because they tell us where things are in the distribution. One of the simplest measures of location that have heard before is the **mode**.

- The **mode** is the most frequent value or category of a variable. It corresponds to peaks on the distribution. The mode tells us what the most likely values in the distribution are. The following

are some key things to consider about the mode

- The mode may not be unique. For example, a bimodal distribution will have two different values for the mode
- The mode is NOT a measure of center. For symmetric distributions with a single peak it will be approximately the same as the median and mean but for distributions that are skewed or have multiple peaks the mode may not be anywhere near the center of the distribution.

## Measures of position: percentiles and quartiles

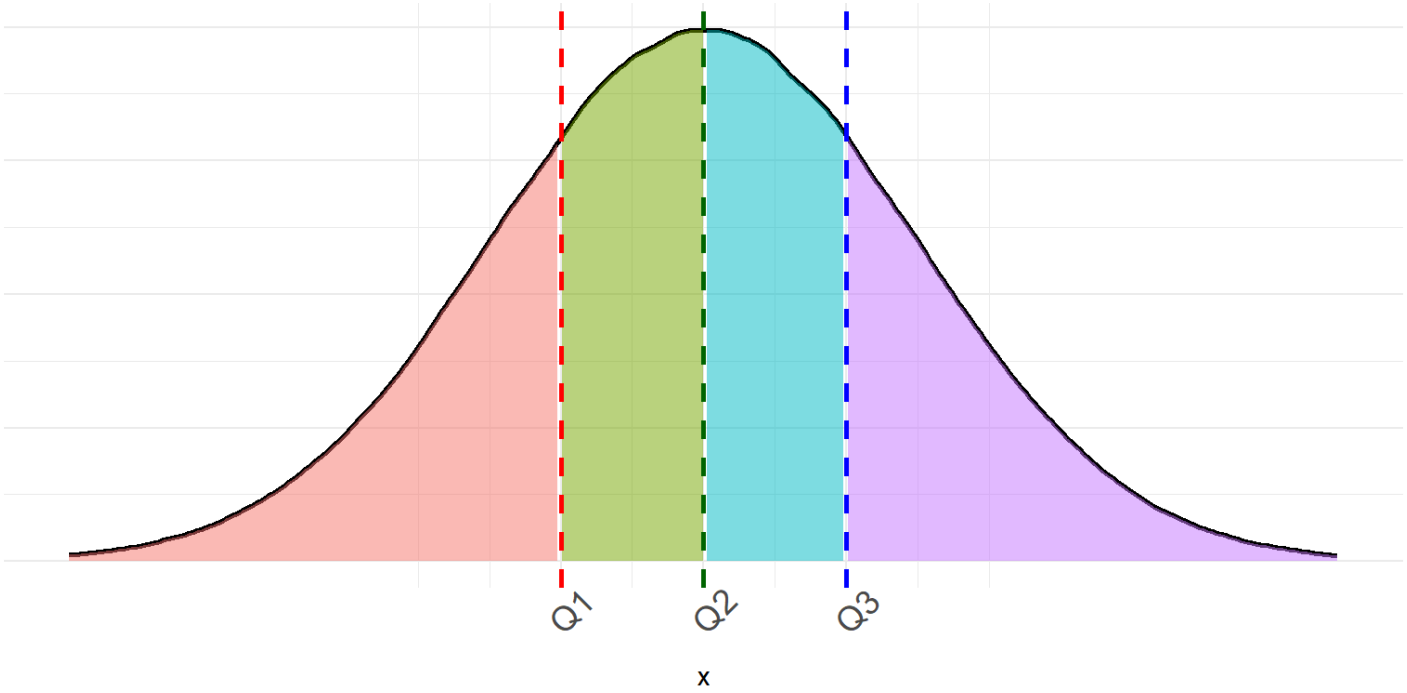
Other useful measures of position are **percentiles**. The $p^{th}$ percentile of a distribution is the value that has $p$ percent of the observations fall at or below it. Take the following simple example of 13 observations of the outcome of rolling a fair six-sided die

$$X = \{1, 2, 2, 2, 2, 2, 3, 4, 4, 5, 5, 6, 6\}$$

What is the $50^{th}$ percentile of $X$? Given the definition above the $50^{th}$ percentile of $X$ is $3$ because half of the observations in the sample are at or below 3. Consequently, the $50th$ percentile is also the median. By definition the median divides the data in equal halves with $50\%$ of observations at or below the median and the remaining $50\%$ of observation having a value greater than or equal to the median.

Two other commonly useful percentiles are the $25^{th}$, and $75^{th}$ percentiles. Together with the median we call these percentiles the quartiles because they divide a distribution into quarters.

- We denote the first quartile ($25^{th}$ percentile) as $Q1$. It is the median of the lower half of the data.
- We denote the $50^{th}$ percentile or median as $Q2$
- We denote the third quartile ($75^{th}$ percentile) as $Q3$. It is the median of the upper half of the data




Example: Consider the following 15 exam scores of students in a statistics class:

$$\{61, 61, 65, 65, 66, 68, 69, 73, 74, 75, 76, 78, 79, 90, 94\}$$

Find the Q1, Q2, and Q3 quartiles for these exam scores. * It's easiest to start with Q2 and work from there. Q2 is the median of the data and, since $n$ is odd will just be the middle value. In this example $Q2 = 73$. Now that we have the middle value we can divide the data into its upper and lower half. The lower half of the data contains the values

$$\{61, 61, 65, 65, 66, 68, 69\}$$

To find the first quartile we simply find the median of this subset of the observations which is $Q1 = 65$. Lastly we repeat this process for the upper half of the data to find the value of the third quartile. The upper half of the exam scores is the subset

$$\{74, 75, 76, 78, 79, 90, 94\}$$

and the median is $78$, therefore $Q3 = 78$.